



HAPTION
Atelier Relais de Soulgé / Ouette
Route de Laval
53 210 SOULGE / OUETTE



15/04/2022

**DESKTOP
INSTRUCTIONS
(TRANSLATION OF THE ORIGINAL INSTRUCTIONS)**

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HAPTION – 8 ZA Route de LAVAL

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S.A à Directoire et Conseil de Surveillance. au capital de 142 240 € – R.C.S. Laval B 439 218 801 – APE 6201Z

This documentation or excerpts thereof may not be reproduced or disclosed to third parties without the express permission of HAPTION S.A.

We have checked the content of this documentation for conformity with the hardware and software described. Nevertheless, discrepancies cannot be precluded, for which reason we are not able to guarantee total conformity.

The information in this documentation is checked on a regular basis, however, and necessary corrections will be incorporated in the subsequent edition.

Subject to technical alterations without an effect on the function.

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 2
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CONTENTS

A. MANUFACTURER.....	3
B. DESIGNATION OF THE MACHINE	3
C. EC DECLARATION OF CONFORMITY	4
D. GENERAL DESCRIPTION OF THE MACHINERY	5
E. DRAWINGS, DIAGRAMS, DESCRIPTIONS AND EXPLANATIONS NECESSARY FOR THE USE, MAINTENANCE AND REPAIR OF THE MACHINERY AND FOR CHECKING ITS CORRECT FUNCTIONING :	6
F. DESCRIPTION OF THE WORKSTATION.....	7
G. INTENDED USE OF THE MACHINERY	8
H. WAYS IN WHICH THE MACHINERY MUST NOT BE USED	8
I. ASSEMBLY, INSTALLATION AND CONNECTION INSTRUCTIONS.....	9
I.1. PREVENTION FROM ASSEMBLY MISTAKES	9
I.2. POWER ON.....	10
I.3. POWER OFF.....	11
I.4. DEVICE STATUS.....	12
I.5. HAPTION CUSTOMER SUPPORT.....	12
J. INSTRUCTIONS FOR REDUCING NOISE OR VIBRATION.....	12
K. PUTTING INTO SERVICE - TRAINING OF OPERATORS	13
L. RESIDUAL RISKS	13
M. PROTECTIVE MEASURES TO BE TAKEN BY THE USER	13
N. TOOLS WHICH MAY BE FITTED TO THE MACHINERY.....	13
O. STABILITY OF THE MACHINERY	13
O.1. FIXATION	13
P. TRANSPORT, HANDLING AND STORAGE OPERATIONS	13
P.1. MANUAL TRANSPORT	14
P.2. PACKAGING PROCEDURE	14
Q. OPERATING METHOD TO BE FOLLOWED IN THE EVENT OF ACCIDENT OR BREAKDOWN	14
Q.1. MOBILE PARTS RELEASE.....	14
R. ADJUSTMENT AND MAINTENANCE OPERATIONS	15
R.1. TYPE AND FREQUENCY OF INSPECTIONS AND MAINTENANCE.....	15
S. INSTRUCTIONS DESIGNED TO ENABLE ADJUSTMENT AND MAINTENANCE.....	15
T. SPARE PARTS TO BE USED	15
U. AIRBORNE NOISE EMISSIONS	15
V. NON-IONISING RADIATION EMITING	15

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 3
---	---	---

A. MANUFACTURER

HAPTION – 8 ZA Route de LAVAL
53210 SOULGE SUR OUETTE – France

B. DESIGNATION OF THE MACHINE

The machine delivered with these instructions is commonly called DESKTOP 3D/6D, and it will be named "Haptic Device" in this document.

Desktop is a haptic interface with 6 degrees of freedom and 3 or 6 active joints, specially designed to interact in Virtual Reality environments.

Its workspace and compact size make it suitable for individual use, for example in front of a desk with a monitor or a Virtual Reality headset.

It finds its place in the following uses:

- Medical simulator
- Design office, feasibility verification
- Behavioral studies (cognitive, functional, illusion) in virtual reality

This manual is intended for the following references:

- o PF3D003321 [F]
- o PF6D003158 [F]



EU DECLARATION OF CONFORMITY

1) Product :

DESKTOP (HAPTIC DEVICE)

CONFIGURATION : ☐ **3D** ☐ **6D**

SERIAL N° :

CUSTOMS CODE (HS CODE) : 84798997

2) Name and address of the manufacturer

HAPTION S.A

Limited company with a capital of 142 240 €

439 218 801 RCS LAVAL

8 ZA Route de LAVAL

53210 SOULGE SUR OUETTE

France

3) This declaration of conformity is issued under the sole responsibility of the manufacturer.

4) Object of the declaration:

DESKTOP

5) The object of the declaration described above is in conformity with



the relevant Union harmonization legislation :

Machinery Directive (2006/42/EC)

RoHS Directive (2011/65/EU)

Low Voltage Directive (2014/35/EU)

ElectroMagnetic Compatibility Directive (2014/30/EU)

6) References to the relevant harmonised standards used :

EN 55022 : 2010 ; EN 55024 : 2010 ; EN 61000-3-2:2006+A1:2009+A2:2009 ; EN 61000-3-3:2008

The technical documentation was created according to Annex VII from the machine directive .The technical information will be transmitted on request duly asked by the national authorities.

The noise level emitted by this machine is below 70dB (A).

Signed for and on behalf of :

at Soulgé-sur-Ouette, on the

François Louveau

President

D. GENERAL DESCRIPTION OF THE MACHINERY



The haptic device is composed of the following components:

1. The haptic arm
2. The external activation sensor
3. The handle
4. Power supply
5. (Configurable function)
6. Double clamping lever
7. (No function)

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 6
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Motors: Type DC - Output power 28 W (joints 1 to 3) and 5 W (joints 4 to 6) in 48V
 Power supply: 240 VAC one-phase
 Translation force: 10 N (maximum), 3 N (continuous)
 Rotation torque (6D configuration): 0,8 Nm (maximum), 0.2 Nm (continuous)
 Maximum control stiffness: 1000 N/m (translation), 4 Nm/rad (6D configuration rotation)
 Apparent inertia: 350 g
 External dimensions: H 400 x L 400 x l 250 mm

Joints amplitude :

Axis 1 to 3 (Base rotation, shoulder and Elbow) : 90°
 Axis 4 (Azimuth - 6D configuration) : 200°
 Axis 5 (Elevation - 6D configuration) : 100°
 Axis 6 (Handle rotation - 6D configuration) : 300°
 Axis 4 (Azimuth - 3D configuration) : 260°
 Axis 5 (Elevation - 3D configuration) : 105°
 Axis 6 (Handle rotation - 3D configuration) : 360°

E. DRAWINGS, DIAGRAMS, DESCRIPTIONS AND EXPLANATIONS NECESSARY FOR THE USE, MAINTENANCE AND REPAIR OF THE MACHINERY AND FOR CHECKING ITS CORRECT FUNCTIONING :

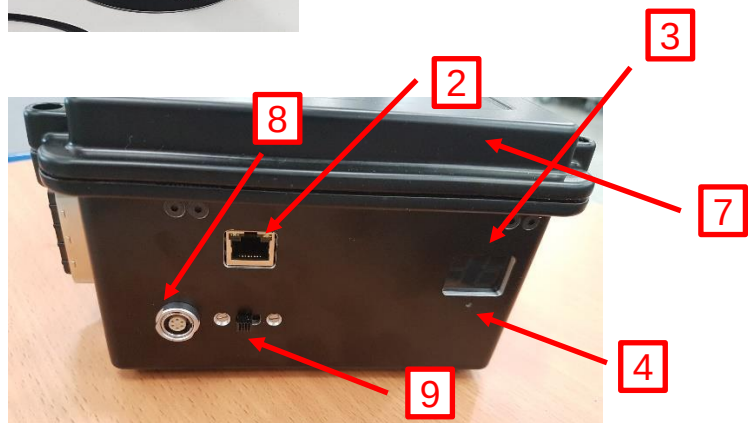
Front of the haptic arm

1) Force Feedback switch



Front of the power supply box

2) Ethernet Interface
 3) Status LED
 4) IP Reset (-> 192.168.100.53)



Back of power supply box

5) Power supply input

6) On/Off Switch

7) Connector for haptic device

8) Connector for the remote
activation sensor

9) "Activation management"
selector



10) Button pad (with activation
sensor)

11) Connection plug of the button
pad



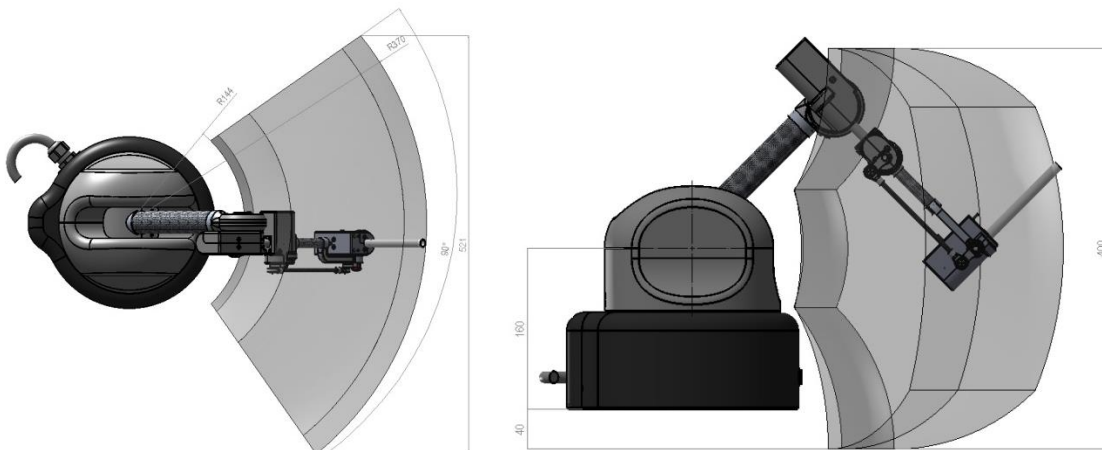
F. DESCRIPTION OF THE WORKSTATION

The user will have to make sure the use of the product provided by Haption does not deteriorate the working conditions nor the protections to the danger he is exposed to.

Only one user must be in the operational workspace described below. The activation sensor shall be installed within range of user's hand. The support for the haptic arm should be chosen carefully, so that the user can reach all points of the operational workspace in a comfortable way.

Around the work envelope of the device, it is important to leave a space free of obstacles, where the user can move freely. A distance of 50 cm is recommended. In case of a fast movement of the device, the free space could be used by the user to escape rapidly without risking encountering an obstacle.

The room lighting should be sufficient in order to see the position of the handle and detect abrupt movements of the device.



	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 8
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G. INTENDED USE OF THE MACHINERY

The work envelope should be free of obstacles, and the user should enter the work envelope with his hand and arm only.



If the haptic device is to be part of a complete system, then the complete system must be CE marked on completion of the installation, including a risk assessment.

The user has to verify every part of the device is firmly attached to the device.

The haptic device is intended for manual interaction through the hand manipulation of the device's handle inside the operational workspace, to perform simulations (interaction with a simulated system).

The possible risks and requirements applying to the systems in which the haptic device is integrated must be taken into account by the integrator.

Use of the device is only permitted under the specified environmental conditions : 5-35°C and relative humidity (non-condensing) from 10% to 90%.

His electromagnetic compatibility compliance is not ensured in residential areas.

The stops in the limits of the workspace withstand the efforts of the motors, corresponding to the specifications of the device.

Applying forces greater than 10 N in translation or 0.8 Nm in rotation may damage the device.

H. WAYS IN WHICH THE MACHINERY MUST NOT BE USED

As delivered, the haptic device must not be used for:

- uses excluded from the scope of directive 2006/42 / EC (machinery directive),
- uses falling within the scope of regulations not included in the certificate of conformity (such as regulation 2017/745/EU relating to medical devices or directive 2014/34/EU relating to devices used in explosive atmospheres, etc.).

Any use or application deviating from the intended use is deemed to be impermissible misuse.

Examples of contraindications to use:

- Operation outside the permissible operating parameters
- Use in potentially explosive environments
- Use the activation sensor as a stop switch



Caution !

Changing the structure of the haptic device, e.g. by drilling holes, attaching external equipment, etc., can result in damage to the components. This is considered improper use and leads to loss of warranty and liability entitlements.



Caution!

When disconnecting cables, pull on the connector body only, never pull on the cables.



I. ASSEMBLY, INSTALLATION AND CONNECTION INSTRUCTIONS



I.1. PREVENTION FROM ASSEMBLY MISTAKES



Caution!

The absence of a proper ground in the power supply can lead to abnormal behaviour and deterioration of the haptic device.

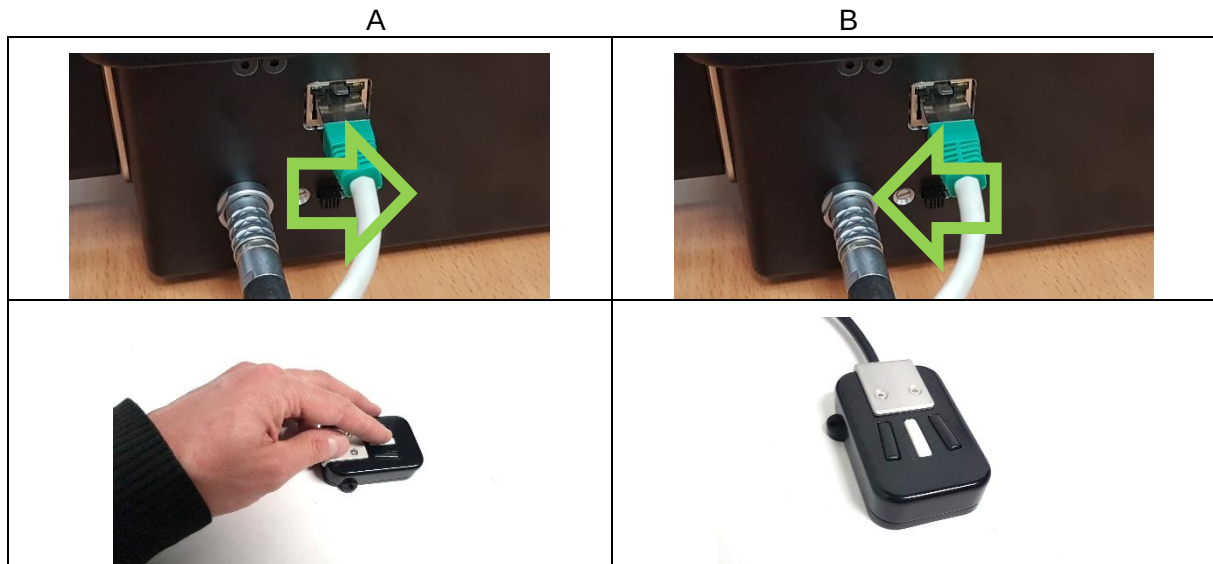
Except the grounding, no electrical protection of the user is integrated into the system. The system must therefore be connected to an installation providing this protection (30 mA/Hi/16A/C-curve differential circuit breaker for Europe).

I.2. POWER ON

1. Connect the power supply box to the mains then connect to this box the power cable of the arm. You should hear a metallic “click” when the plug is in the correct position.



2. Connect the plug (11) to the connector (8)



Mode A (selector 9 to the right): Force feedback can only be activated if the hand is placed on the activation sensor of the button pad. This mode is recommended for controlling risks (software tests, sensitive operations, etc.)

Mode B (selector 9 to the left): Operation without activation sensor. It is possible to unplug the button pad.

3. Connect the Ethernet/Ethercat interface (2) to your PC.



For optimal performance, you should use a shielded Cat5e FTP cross-over RJ45 cable of the correct length, directly connected to an Ethernet port of your workstation. USB/Ethernet adapters are not recommended, as they do not support full-duplex communications.

In case it is not possible to have a direct connection to the workstation, then add a 1 Gbit/s Ethernet hub immediately behind your workstation, and use a direct (not cross-over) shielded RJ45 cable.

The default network settings of the haptic device are:

- IP address: 192.168.100.53
- UDP port: 5000

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 11
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On Linux, to execute steps 4 to 7, follow the instructions in the "LINUX SUPPORT" sheet of the "Virtuose maintenance - Collection of sheets" document.

4. Communication protocol configuration

If the haptic device has an EtherCAT interface, then install the EtherCAT driver (see Virtuose maintenance -Collection of sheets) before installing the controller service.

Whatever the protocol (Ethernet or EtherCAT), install the controller service (SvcHaptic_...) by running the program "HAPTIC_SERVER..._setup.exe". The default software port proposed for the connection to the application (API - Controller Service) is 53210.

With Ethernet protocol :

At delivery, the IP address for the haptic device is 192.168.100.53. Check with your network administrator that this setting and the software port are compatible with your IT environment and do not conflict (including the settings of the haptic device). On the other hand :

- Validate these settings anyway and complete the services installation
- Set up the network card connected to the Haption hardware, to make it compatible with the default IP addressing
- Launch the "HapticConfigurator.exe" program and modify the addresses to make them compatible with your environment
- Restore the configuration of the network card connected to Haption hardware
- Restart the computer connected to the Haption hardware

5. Check that the power button is not pressed.

Switch the power switch to ON.

The status 00 should be displayed after a few seconds.

6. Calibration :

- Launch the program : Windows Start Menu/All Programs/HAPTION Haptic Server Virtuose.../Calibration!
- A Windows console appears
- If the message "ERROR: problem of communication" appears, follow the maintenance instructions of the document "Troubleshooting Communication"!
- When instructed by the software : press the power button!
- The arm moves to its calibration positions and then stops
- When instructed by the software : depress the power button!
- In the end, the displayed status is 71

7. Control :

- Copy the USB key's folder VirtuoseDemo on the desktop
- Start "start_demo"
- The "Virtuose Demo" window opens
- Press the power button. If it does not light, close the program VirtuoseDemo
- Use :
 - o The left button to select an object
 - o The middle button to shift the arm when it is in abutment
 - o The right button to activate the virtual guide
- Before closing the program, release the power by pressing the power button
- To close the program, put the cursor in the demo window, then press Esc

I.3. POWER OFF

Power OFF the switch (6)

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 12
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I.4. DEVICE STATUS

The status of the device is indicated by the display on the back of the power supply unit.

If the system is out of service, then it can only be restarted after the power has been switched off and the issue fixed.

In case the force feedback switches off, then it can be reactivated by pressing the power button to release it and then pressing it again.

00: No failure, awaiting power authorization from the software

IP: Reset the factory setting (press the reset button when booting haptic device)

11-16: Error DSP joint 1-6 - System is out of service

21-26: Error encoder joint 1-6 - System is out of service

31-36: error signal joint 1-6 - System is out of service

41-46: Alarm T ° joint PWM 1-6 - System is out of service

51-56: PWM error joint 1-6 - System is out of service

70: Relay error - System is out of service

71: Interruption of network communication (end of calibration ...) - The force feedback is switched off

72 : The emergency stop button is pressed or unplugged

88 : System OK

91-96: Force feedback restricted to avoid a risk of overheating of the motor of axis 1-6

I.5. HAPTION CUSTOMER SUPPORT

The HAPTION documentation offers information on operation and provides assistance with troubleshooting.

For further assistance, please contact your local HAPTION reseller or the HAPTION hotline.

The following information is required for processing a support request:

- Model and serial number of the haptic device
- Type of software used with the haptic device
- Version of the software used with the haptic device
- Description of the problem, duration and frequency of the fault

The list of local HAPTION resellers is available on our website www.haption.com, under the tag "Partners".

In case no local reseller can be found in your country, you should contact HAPTION directly by using one of the support e-mail addresses.

1. For other software issues: software@haption.com
2. For sales and marketing: sales@haption.com
3. For general questions: contact@haption.com

In case of urgent need, or if you receive no answer to your e-mail, you can call the HAPTION hotline, at office hours Paris time :

Phone number: +33 (0)2 43 64 51 20

J. INSTRUCTIONS FOR REDUCING NOISE OR VIBRATION

None

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 13
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K. PUTTING INTO SERVICE - TRAINING OF OPERATORS

The haptic device cannot be used without installation of the controller software. Refer to the installation instructions for details on the procedure.

The only form of training needed for the use of the device is the knowledge and application of the instructions contained in this manual.

It is recommended to check the operator's knowledge of :

- calibration
- shutting down the system

L. RESIDUAL RISKS

In a normal use, the haptic device does not present risks requiring the prevention implementation other than those described in those instructions.

M. PROTECTIVE MEASURES TO BE TAKEN BY THE USER

When it is necessary to prevent biological contamination by contact, the parts of the haptic device handled by the user can be disinfected with a cloth soaked in hydroalcoholic solution, wrung out sufficiently so as not to cause sagging. The contact time between the haptic device and the hydroalcoholic solution should not exceed one minute.

N. TOOLS WHICH MAY BE FITTED TO THE MACHINERY

We guarantee the normal use of the machine only through the tools provided by the company. The use of any accessory not provided by Haption must be done under the instructions of the directive 2006/42/CE

O. STABILITY OF THE MACHINERY

O.1. FIXATION

It is advisable to fix the haptic device's base on a rigid support. The central hole for M6 screw on the underside of the haptic device allows to do it.



Warning!

If not fixed rigidly on a stable support, the haptic arm might fall over during operations, thus causing injury to personal and damage to equipment in the vicinity.

P. TRANSPORT, HANDLING AND STORAGE OPERATIONS

Weight of the haptic arm : 3.6 kg

Weight of the power supply : 2.1 kg

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 14
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Environmental conditions for transport and storage:

- Temperature : from -10°C to +80°C
- Hygrometry: from 10% to 90% without condensation

P.1. MANUAL TRANSPORT

The haptic device can be transported by one adult person without special equipment.

P.2. PACKAGING PROCEDURE

The haptic device must be packed in a rigid container. It must be propped up with a supply filling allowing to avoid any shock between the container and the haptic device.



Q. OPERATING METHOD TO BE FOLLOWED IN THE EVENT OF ACCIDENT OR BREAKDOWN

Q.1. MOBILE PARTS RELEASE

No particular danger exists for the user when releasing the mobile parts.

	INSTRUCTIONS DESKTOP	Notice Desktop3D6D MEP11.00 TXT9.01.docm Page 15
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R. ADJUSTMENT AND MAINTENANCE OPERATIONS

R.1. TYPE AND FREQUENCY OF INSPECTIONS AND MAINTENANCE

FREQUENCY	MAINTENANCE, INSPECTIONS	CRITERIA OF CHANGE OF PART
Every 1000h to 2000h	Changing transmission cables, confer to maintenance instructions. Changing weight-compensation cables, confer to maintenance instructions.	Lack of tension of the springs Damage to the cable plastic coating
At least 1 / 500 hours of operation At least 1 / 6 calendar months	Tactile and visual inspection Check of the activation sensor	

S. INSTRUCTIONS DESIGNED TO ENABLE ADJUSTMENT AND MAINTENANCE

To operate a maintenance task on the device, you have to put yourself in contact with Haption, who will transfer you specific instructions for the maintenance of your device.

T. SPARE PARTS TO BE USED

The only elements to be used are those provided by Haption.

U. AIRBORNE NOISE EMISSIONS

The acoustic power level sent and A level-headed to workspaces is 70dB (A). The maximum value of acoustic pressure instantly sent and C level-headed to workspaces is below 63Pa (130dB compared to 20 μ Pa).

V. NON-IONISING RADIATION EMITING

The machine can't emit non-ionizing radiation capable of harming people, especially people wearing active or non-active implantable medical devices.